

GymTrack

Gym Membership Management System

Phase 3 — Development Progress & Feature Integration Report

Project Title	GymTrack — Gym Membership Management System
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Course / Section	IT342-G01 — Systems Integration and Architecture 1
Institution	Cebu Institute of Technology — University (CIT-U)
Date	July 15, 2026
GitHub Repository	https://github.com/louisflim/gymtrack

Repository link: <https://github.com/louisflim/gymtrack>

1. Development Progress Summary

GymTrack is a full-stack gym membership system with a Spring Boot backend, React web client, and Android mobile app. Early work established authentication, dashboards, plans, members, memberships, PayMongo payments, and QR attendance. Mid-phase work reorganized the codebase into vertical feature slices (auth, plans, members, membership, payments, attendance, staff, dashboard) shared across web and mobile. Recent work focused on account security, safer admin member removal, production-ready API hardening, clearer errors, and a shared Settings experience for all roles.

Progress milestones:

- Backend + web + mobile foundations (auth, gym model, dashboards).
- Feature vertical slices for registration, login, staff creation, plans, members, membership, payments, attendance, and admin KPIs.
- PayMongo checkout integration (mock/test-friendly for school deployment).
- QR enrollment and attendance check-in/check-out for staff and members.
- Staff forced password change on first login; reusable change-password API.
- API method security and production secret/profile guards.
- Admin “remove member from gym” (unenroll) instead of hard-deleting the user account.
- Settings menu (web + mobile): change password and self-delete account.

2. Newly Implemented Features

2.1 Account Settings (Web & Mobile)

A Settings tab is available to Admin, Staff, and Member. Users can change their password and permanently delete their own account (password confirmation + confirm dialog). Light/dark theme toggles were

intentionally not included. Shared components keep web and mobile behavior aligned (ChangePasswordForm / SettingsPanel on web; layout_settings_panel + SettingsViewModel on mobile).

2.2 Staff First-Login Password Reset

When an admin creates a staff account, mustChangePassword is set. Staff must set a new password via POST /api/auth/change-password before using the dashboard. Web and mobile redirect accordingly.

2.3 Soft Remove Member from Gym

Gym owners remove members from their gym without deleting the users row. Memberships are expired, gym_id is cleared, and the login account remains so the person can join again later. Confirm dialogs on web and mobile clarify this behavior.

2.4 Security Hardening for Deployment

Added @EnableMethodSecurity, @PreAuthorize on controllers, HTTP role matchers, production profile validation for secrets, CORS/webhook guards, and safer PayMongo mock ownership checks.

2.5 UX Reliability Fixes

Fixed login/register buttons that stayed disabled until re-render, staff login after password change, and friendlier API error messages across clients.

3. How Features Were Integrated

Integration follows a shared API contract: React (axios) and Android (Retrofit) call the same Spring Boot endpoints under /api/*. Session tokens (JWT) are stored in localStorage (web) and DataStore (mobile). Feature modules are organized as vertical slices so UI, repositories, and backend packages stay aligned (e.g., feature/auth, feature/settings, feature/members).

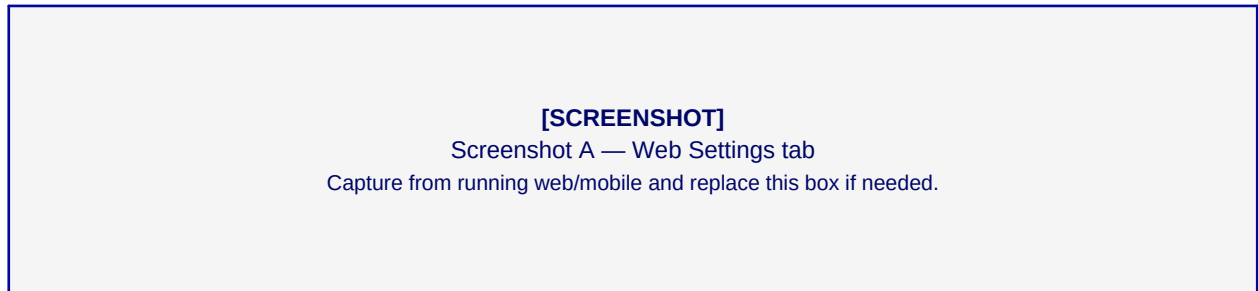
- Settings → POST /api/auth/change-password (existing) and DELETE /api/auth/account (new AccountService).
- Admin member remove → DELETE /api/members/{id} now unenrolls (MemberService) instead of hard delete.
- Staff password gate → AuthResponse.mustChangePassword drives /change-password and ChangePasswordFragment.
- Nav config synced: web dashboardUi.js ↔ mobile DashboardUiCopy.kt (including Settings tab).
- Reusable UI: ConfirmDialog (web), MaterialAlertDialogBuilder (mobile), shared password form on web.

4. Screenshots of Working Features

Capture these views while the backend, web, and/or Android app are running. Placeholder frames below mark the required screens for the submission package.

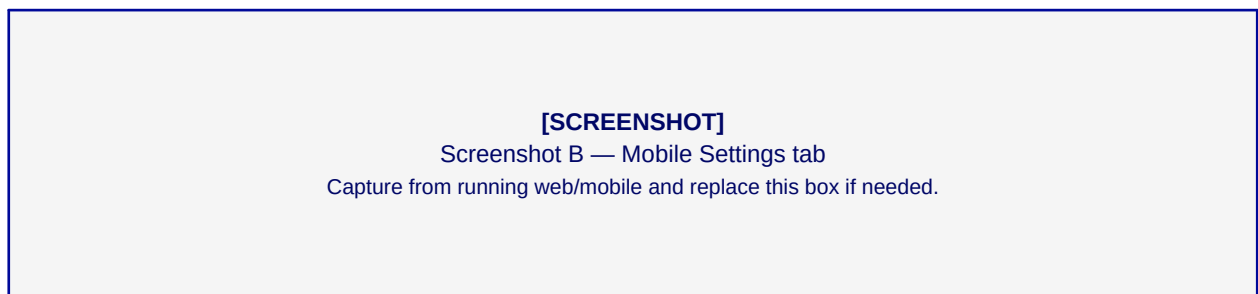
Screenshot A — Web Settings tab

Dashboard → Settings: Change Password form and Delete Account section (all roles).



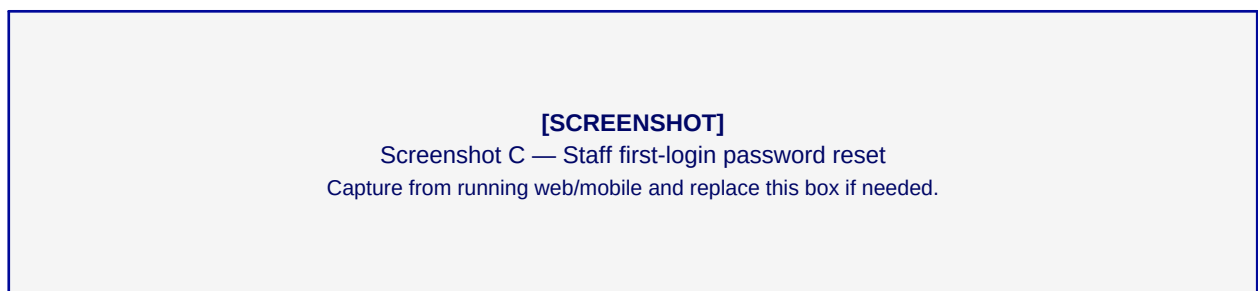
Screenshot B — Mobile Settings tab

Android dashboard Settings panel with password fields and Delete Account action.



Screenshot C — Staff first-login password reset

Forced reset password screen after staff signs in with a temporary password.



Screenshot D — Admin remove member confirmation

Confirm dialog stating the member is removed from the gym; account is not deleted from the database.

[SCREENSHOT]

Screenshot D — Admin remove member confirmation
Capture from running web/mobile and replace this box if needed.

Screenshot E — Member QR / attendance (integrated core flow)

Member QR code or staff scan flow confirming attendance integration still works.

[SCREENSHOT]

Screenshot E — Member QR / attendance (integrated core flow)
Capture from running web/mobile and replace this box if needed.

5. End-to-End Workflow Test Results

Test #	Workflow	Steps	Expected	Result
E2E-1	Staff first login → change password → dashboard	Admin creates staff → staff logs in → forced password page → sets new password → dashboard loads → can log in again with new password	mustChangePassword cleared; JWT session continues; no login loop	PASSED
E2E-2	Admin removes member from gym (soft unenroll)	Admin opens Members → Remove → confirm → list refreshes → check users table for same email	Member disappears from gym list; users row remains with gym_id NULL; memberships expired	PASSED
E2E-3	Settings: change password then delete account	Any role → Settings → update password → sign in again → Settings → enter password → Delete Account → confirm	Password updated successfully; after delete, login fails and related personal records are removed	PASSED
E2E-4 (bonus)	Member PayMongo / mock pay → membership active → QR check-in	Member enrolls via gym QR → chooses plan → completes checkout (mock/test) → shows member QR → staff scans	Membership ACTIVE; attendance log created	PASSED (mock/test mode)

6. Integration Issues Encountered and Solutions

Issue	Cause	Solution
Admin “delete member” wiped the user from the database	DELETE /api/members/{id} called a hard-delete helper; also old Spring Boot process kept serving previous bytecode	Changed MemberService to unenroll (clear gym, expire memberships). Restart backend after code changes. Soft-remove confirmed in users table.
Login / Create Account buttons stayed disabled while typing	Enabled state only updated when ViewModel uiState emitted, not on text change	Added local updateButtonState() on doAfterTextChanged for login and register (mobile).

Staff could not log in smoothly after first password change	mustChangePassword / session routing edge cases after reset	Standardized AuthResponse flag handling and post-change navigation to dashboard; regression fix commit b5d441d.
Deploy security risk (open routes / weak secrets)	Role checks incomplete; prod secrets not validated	Added method security, HTTP matchers, application-prod profile validator, webhook signature requirement.
Web & mobile Settings needed identical behavior without theme complexity	Risk of duplicating password/delete logic per platform	Reusable ChangePasswordForm + SettingsPanel (web); SettingsRepository reuses ChangePasswordRepository (mobile); shared /api/auth endpoints.

7. Commit History Table

Selected commits mapped to integration work. Full history is on GitHub main branch.

Feature / Change	GitHub Commit Link
User Settings: change password & delete account (web + mobile)	https://github.com/louisflim/gymtrack/commit/180a35b
Admin remove member from gym (soft unenroll, confirm dialog)	https://github.com/louisflim/gymtrack/commit/315b9a3
Fix staff login after forced first-login password change	https://github.com/louisflim/gymtrack/commit/b5d441d
Security: API role hardening and production secrets	https://github.com/louisflim/gymtrack/commit/2ff61af
Staff first-login password change (mustChangePassword)	https://github.com/louisflim/gymtrack/commit/bbcb15c
User-friendly error messages (web, mobile, backend)	https://github.com/louisflim/gymtrack/commit/1f77a7e
Fix mobile register Create Account button enable state	https://github.com/louisflim/gymtrack/commit/5ffd37
Admin member/staff delete option & PayMongo fixes	https://github.com/louisflim/gymtrack/commit/d3f259d
QR code bugfix and UI updates	https://github.com/louisflim/gymtrack/commit/be3010b
Fix Sign In button not enabling on login	https://github.com/louisflim/gymtrack/commit/6f58dfb
Gym entity bug fixes	https://github.com/louisflim/gymtrack/commit/44e7acb
Phase 2 documentation PDF	https://github.com/louisflim/gymtrack/commit/4088f3b
Vertical slice: staff accounts	https://github.com/louisflim/gymtrack/commit/65f2c36
Vertical slice: admin dashboard KPIs	https://github.com/louisflim/gymtrack/commit/61cd3ea
Vertical slice: QR attendance check-in/out	https://github.com/louisflim/gymtrack/commit/7740660
Vertical slice: PayMongo checkout payments	https://github.com/louisflim/gymtrack/commit/f7f26f7
Vertical slice: membership status	https://github.com/louisflim/gymtrack/commit/8ac7024
Vertical slice: member management	https://github.com/louisflim/gymtrack/commit/887c770
Vertical slice: plan management	https://github.com/louisflim/gymtrack/commit/672bfbf
Vertical slice: admin staff creation	https://github.com/louisflim/gymtrack/commit/3407f40
Vertical slice: user login	https://github.com/louisflim/gymtrack/commit/544bab8
Vertical slice: user registration	https://github.com/louisflim/gymtrack/commit/e87ed8d
Layout and UI updates	https://github.com/louisflim/gymtrack/commit/0e3151b
Mobile login and register	https://github.com/louisflim/gymtrack/commit/d47a6ce
Initialize Android mobile app	https://github.com/louisflim/gymtrack/commit/38ec787
Admin/member dashboard	https://github.com/louisflim/gymtrack/commit/6b9b568
Project README	https://github.com/louisflim/gymtrack/commit/2b11064
Initial Commit	https://github.com/louisflim/gymtrack/commit/b9449f6

8. Individual Contribution Statement

I, **Louis Francis Lim**, am the sole developer of GymTrack for IT342-G01. I designed and implemented the Spring Boot backend, React web application, and Android mobile client; integrated PayMongo payments and QR attendance; refactored features into vertical slices; added staff first-login password change, API security hardening, soft member removal from gyms, and the Settings menus for password change and account deletion on both platforms. All commits in the louisflim/gymtrack repository reflect my individual work unless otherwise noted.

Signature over Printed Name

Date: July 15, 2026